Assignment 4

Digital Signal Analysis
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Due date: 11:59pm on 26th April 2026

1. Solve all the questions and submit a handwritten document.
2. Plagiarism will be penalised.
3. You can use any coding/programming language to complete 'Coding' questions.
4. Submit a pdf of the form `<roll no>_dsa.pdf`
5. The submission should be a zip folder named with your rollnumber-dsa . Each coding question must be placed in a separate folder containing the corresponding code and results.

Questions

1. **(Coding)** Design an FIR linear-phase, digital filter approximating the ideal frequency response

$$H_d(\omega) = \begin{cases} 1, & \text{for } |\omega| \le \frac{\pi}{6} \\ 0, & \text{for } \frac{\pi}{6} < |\omega| \le \pi \end{cases}$$

   (a) Determine the coefficients of a 25-tap filter based on the window method with a rectangular window.

   (b) Determine and plot the magnitude and phase response of the filter.

   (c) Repeat parts (a) and (b) using the Hamming window.

   (d) Repeat parts (a) and (b) using a Bartlett window.

2. Repeat Problem 1 for a bandstop filter having the ideal response

$$H_d(\omega) = \begin{cases} 1, & \text{for } |\omega| \le \frac{\pi}{6} \\ 0, & \text{for } \frac{\pi}{6} < |\omega| < \frac{\pi}{3} \\ 1, & \text{for } \frac{\pi}{3} \le |\omega| \le \pi \end{cases}$$

3. **(Coding)** Redesign the filter of Problem 2 using the Hanning and Blackman windows.

4. Determine the unit sample response $\{h(n)\}$ of a linear-phase FIR filter of length $M = 4$ for which the frequency response at $\omega = 0$ and $\omega = \frac{\pi}{2}$ is specified as
$$H_r(0) = 1, \qquad H_r\left(\frac{\pi}{2}\right) = \frac{1}{2}.$$